Arrays

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Quiz 1: Arrays

1. Which of the following is false?

```
int a[10] ;

int main() {
}
```

What can we say about the array a?

a) All elements will be initialized to 0

b) All elements will have the same garbage value

c) Each element will have a distinct garbage value

d) Results in a syntax error

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Quiz 2: Arrays

2. Comment :

```
int a[] = {10, 20, 30, 40, 50}; int *p = a + 2;

int x = *p++;
```

What happens?

a) x becomes 31; p is not changed

b) x becomes 30; p points to a[3];

c) Syntax error; initialize of p not correct

d) x becomes 40; p points to a[3];

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Quiz 3: Arrays

3. Comment :

```
int a[] = {1, 2, 3}; int b[] = {1, 2, 3};

a == b
```

a) Is a syntax error; arrays cannot be compared

b) 1 as both are arrays contain the same elements

c) 0 as both the arrays point to different locations

d) None of these

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Quiz 4: Functions, Pointers and Arrays

4. Comment :

sizeof int on this implementation is say 4 bytes.

```
int a[] = {10, 20, 30, 40, 50}; int *p = a + 5;
```

what is the value of p – a?

a) 20

b) 5

c) Depends on implementation

d) Error: pointer – array not allowed

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Quiz 5: Functions

5.

int a[5][4]; The array is arranged linearly in the memory of the computer.

Counting from 0, what is the position of a[3][2] in the array?

a) 14

b) 17

c) 13

d) None of these

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Quiz 5: Functions

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c) 13

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Revision: Array

- has # of elements
- all the elements should be of the same type - homogeneous
- size of the array is specified at the point of creation
    - size is given as a constant at compile time
    - size could be specified at runtime in C99 VLA : variable length array
- array cannot grow or shrink
- elements are selected based on index or subscript
- index is an integer; starting value is 0 - zero based indexing
- supports random access
    - time required to access the element does not depend on its position
    - is always same

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Revision Array

- Array at compile time is an array. It knows everything about itself.
- Array at runtime degenerates to a pointer.
    - At runtime, the size of the array is not stored.
- Index out of bounds - accessing elements beyond the size of the array cannot be checked.
    - Checking would require the size of the array at runtime - extra memory
    - Checking would incur some runtime cost.
    - This if done would be penalizing the right programmer
- So, 'C' follows this philosophy. The programmer gets what he deserves!
- C array at runtime is a constant pointer
- So, arrays cannot be assigned one to another. Arrays are not assignment compatible.

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Revision: Array

- Every element of an array itself can be an array
- This is called multidimensional array
- Stored in a row major fashion
- Size of each dimension but for the first required to access an element of the multidimensional array
- If a 2 D array is passed as an argument to a function, the corresponding parameter is a pointer to an array
- void foo(int a[][10]); // only 2 D array having 10 columns can be passed as argument
- void bar(int n, int a[][n]); // Any 2 D array can be passed as argument as long as the first argument indicates the number of columns

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THANK YOU

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